Does the Visual World Paradigm Show Evidence of Referential Adaptation?

Comparing Referential Adaptation Across Two Eye-Tracking Experiments

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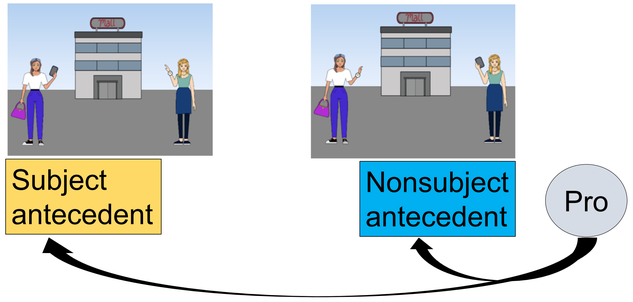
Background

- Ambiguous pronoun resolution is guided by the **subject bias**: listeners prefer to interpret pronouns as referring to the grammatical subject of the preceding clause (Arnold et al., 2000; Jarvikivi et al., 2005; Kaiser, 2009).

  Liz went shopping with Ana at the new mall. She bought a new watch because her old one was broken



- Recent behavioral work shows that comprehenders can **modulate this bias** when exposed to pronoun-referent patterns that violate their expectations — a phenomenon called ***referential adaptation*** (Johnson & Arnold, 2023; Ye & Arnold, 2023; Roy et al., 2025).
- However, **it remains unclear** whether adaptation effects emerge during ***online processing*** or only influence offline judgments and final interpretations
- We report two **Visual World eye-tracking experiments** examining online evidence for referential adaptation, and how **trial structure** affects detectability of adaptation effects.

Research Question

- *Can referential adaptation be detected during online pronoun processing via eye-tracking?*
- *Does trial structure (quasi-blocked vs. pseudo-randomized) modulate the strength of adaptation effects?*

Methods & Design

Design: *Between-subjects* Exposure (subject-referent vs. nonsubject-referent pronouns in 28 items) × *Within-subjects* Target Type (subject vs. nonsubject targets in 16 critical items). There were 4 lists.

Critical trials: Participants heard stories about two same-gender characters; Only one character matched the disambiguating information.

Liz went shopping with Ana at the new mall. She bought a new watch because her old one was broken.

Measure: Empirical logit-transformed (Elogit) fixations to the subject character (vs. all other locations) during the ambiguous period (pronoun onset to 300ms post-disambiguation).

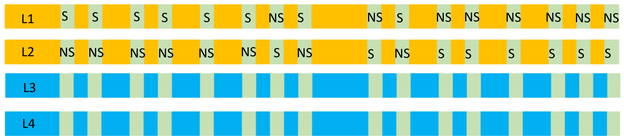
Exposure trials:

Subject exposure story: Will won a trophy with Liz at the spelling bee competition. He was very proud for spelling all the words correctly.

Nonsubject exposure story: Will won a trophy with Liz at the spelling bee competition. She was very proud for spelling all the words correctly

Exp 1 vs. Exp 2:

Experiment 1: Quasi-blocked, first & second halves differed systematically in Target (S vs NS) type.



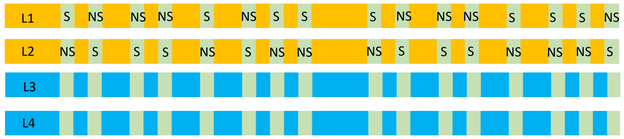
Subject exposure Item: The pronoun in the second sentence always referred to the subject of the previous sentence

Nonsubject exposure Item: The pronoun always referred to the nonsubject of the previous sentence in the story

Critical Item: The pronoun in the second sentence of the story was always ambiguous. target is subject vs. nonsubject

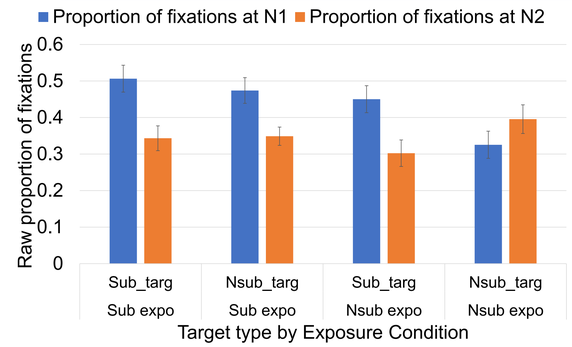
Experiment 2: Improved **pseudo-randomization**

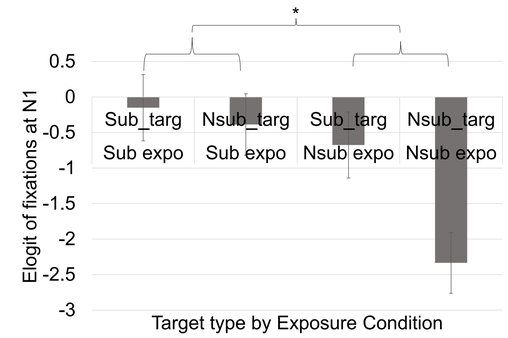
All target types distributed evenly throughout.

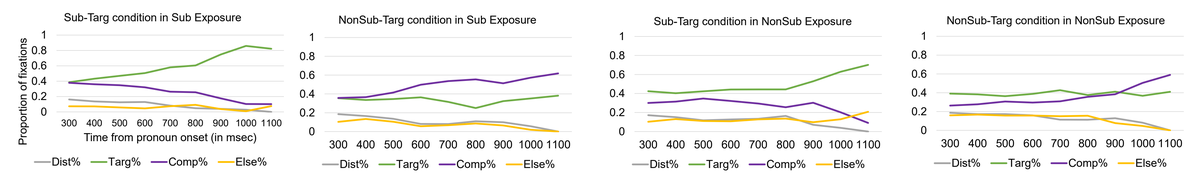


Experiment 1 Results

Key finding: Significant main effect of exposure **F(1, 17.8) = 5.04, p = .038**. Nonsubject-exposure participants showed reduced subject bias overall. No significant interaction between exposure and target type, **F(1, 14.7) = 2.97, p = .106.**

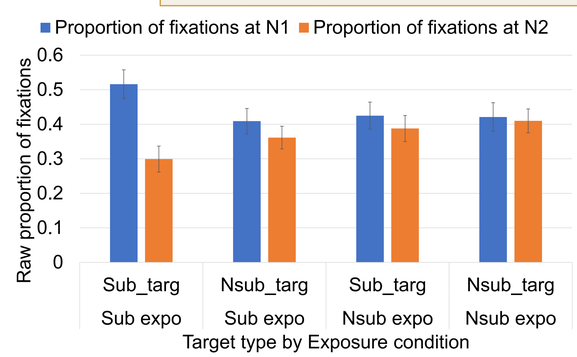


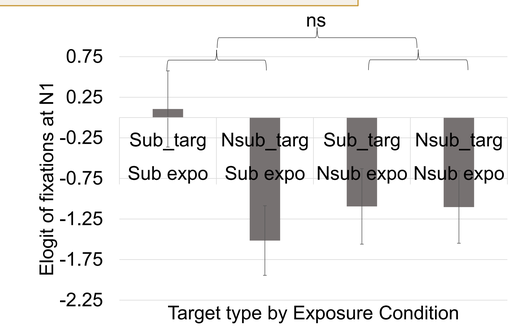


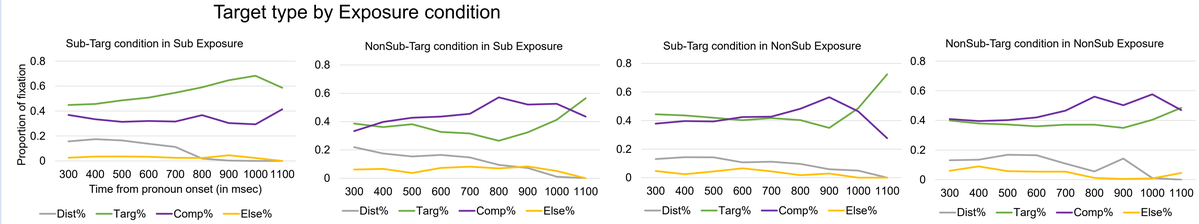


Experiment 2 Results

Key finding: No significant main effect **F(1, 20.6) = 0.43, p = .517**, no significant interaction **F(1, 10.6) = 2.97, p = .114**. But numerical pattern similar to Exp 1 (same F-value).







Discussions

- **Interpretation 1 [Cumulative adaptation]:** The quasi-blocked structure in Exp 1 may have allowed adaptation to build across successive trials, making a small main effect detectable that was too weak to emerge under distributed presentation in Exp 2.
- **Interpretation 2 [Artifactual main effect]:** The main effect in Exp 1 may be an artifact of the inadvertent blocking rather than a genuine adaptation signal.
- **Consistent interaction pattern:** Identical F = 2.97 for interaction in both Exps. suggests potential modulation of adaptation by target type, but larger samples are needed to detect this reliably.

Conclusions

Mixed evidence for online referential adaptation: a main effect emerges under quasi-blocked presentation but not under pseudo-randomized presentation.

Findings highlight the importance of trial ordering and statistical power when studying incremental adaptation in online processing.

References

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